Project code

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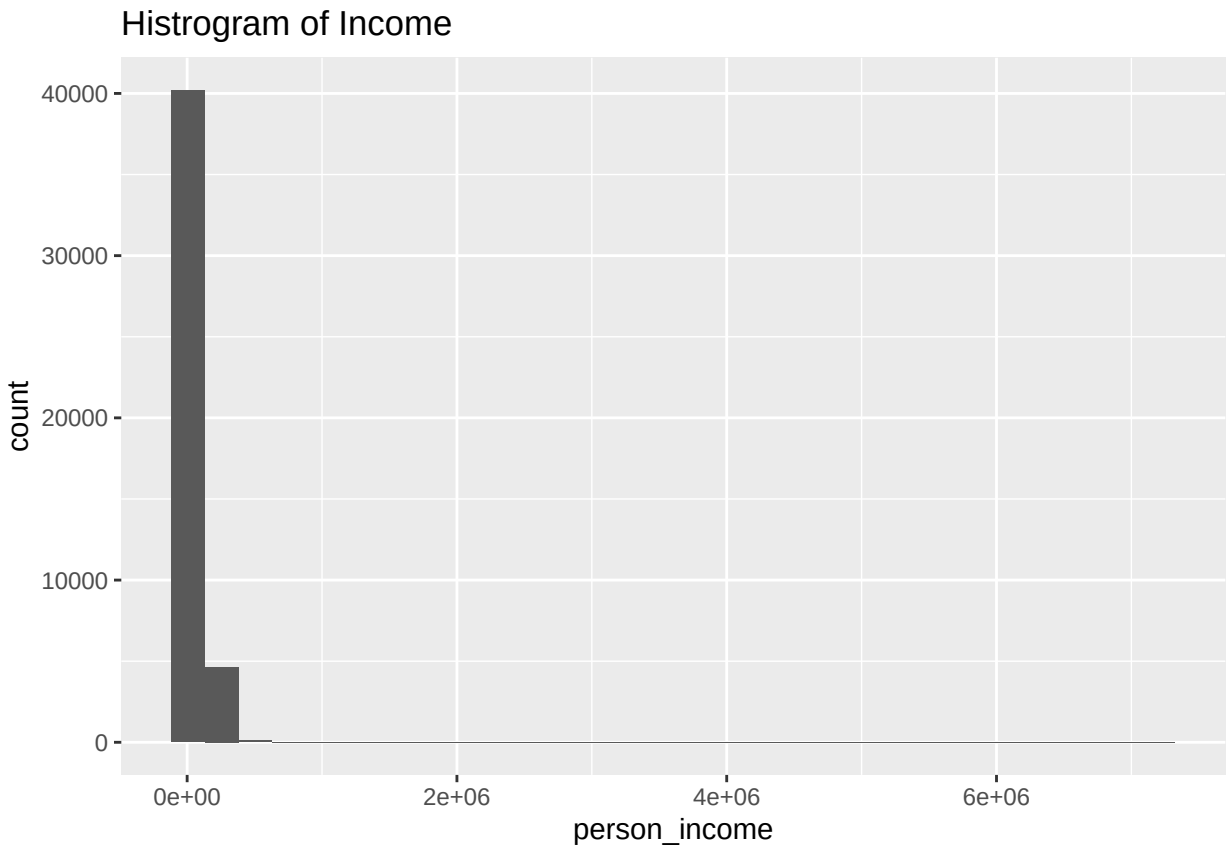
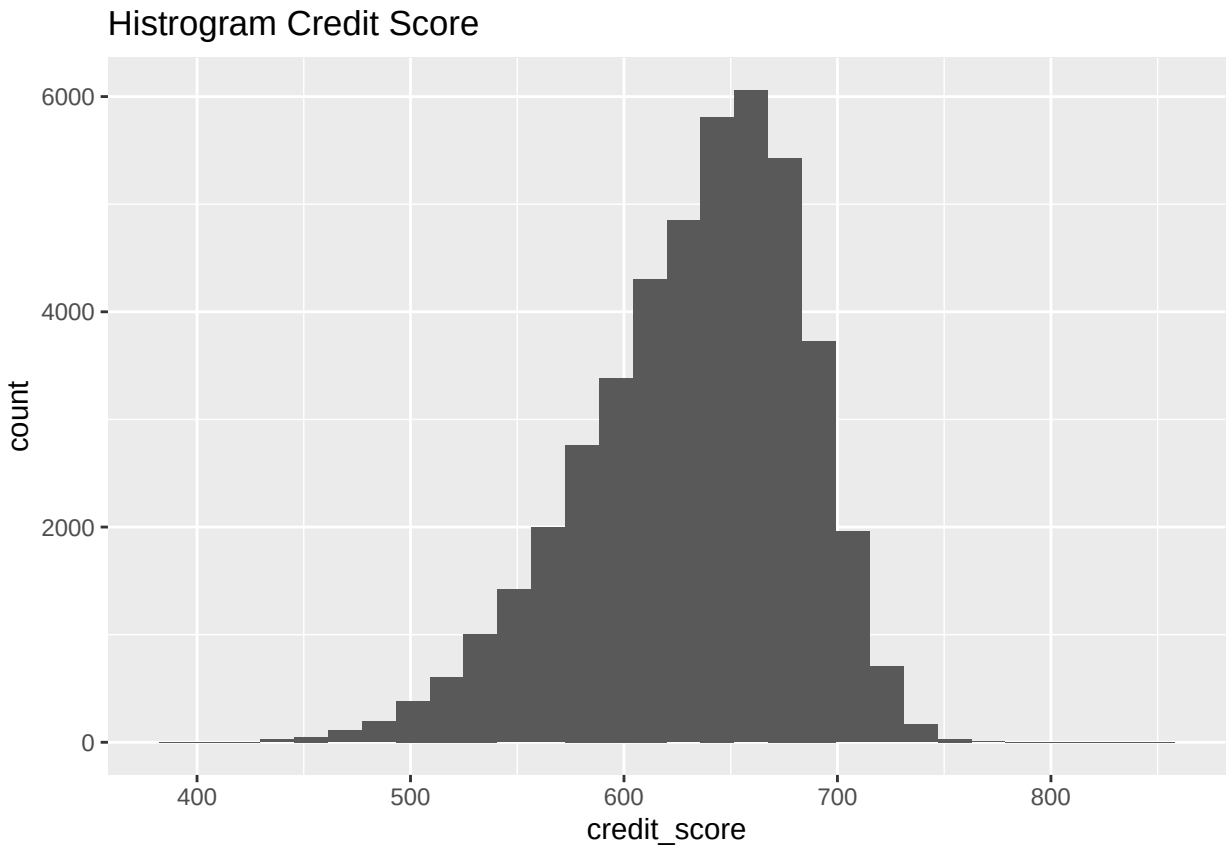
Univarite Data

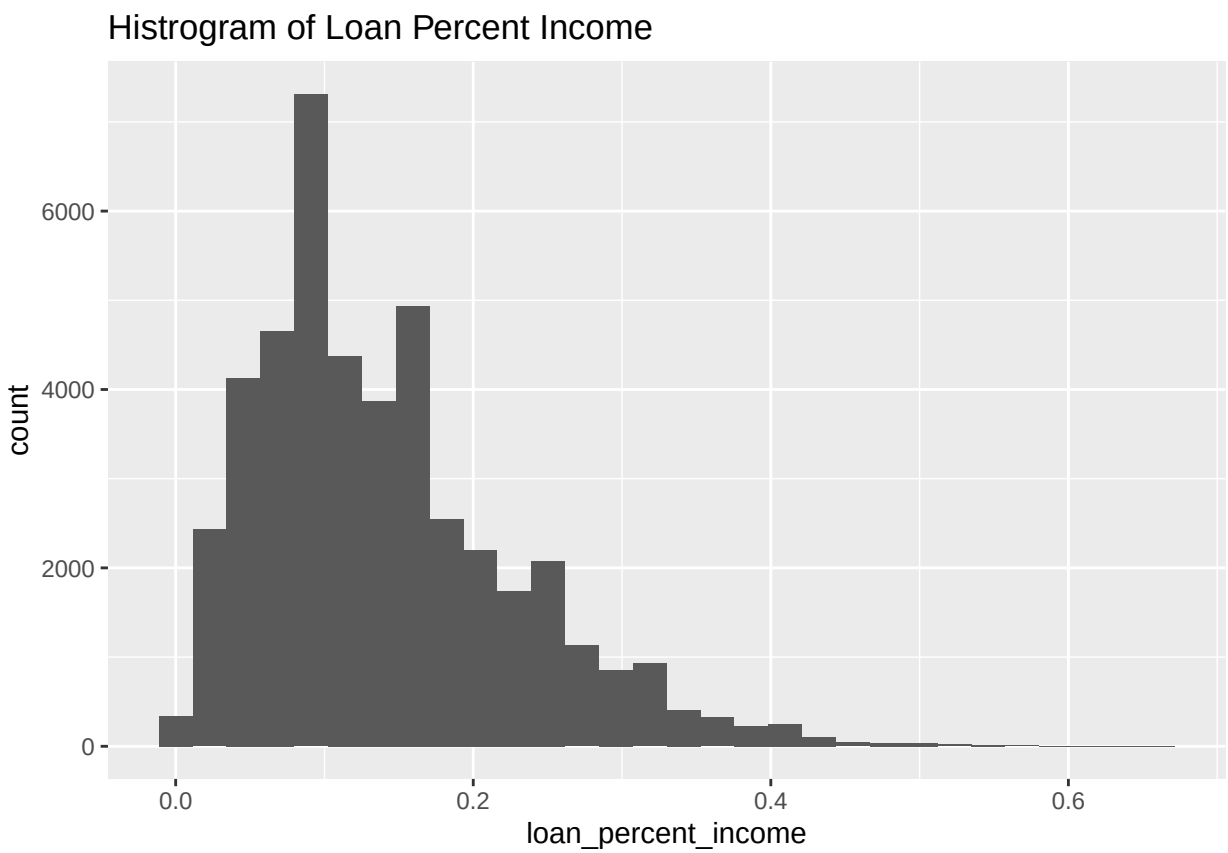
```
loan <- read.csv("/Users/seandasovich/Downloads/loan_data.csv")
library(ggplot2)

loan$loan_status <- factor(loan$loan_status, levels = c(0, 1),
                           labels = c("Rejected", "Approved"))

loan$previous_loan_defaults_on_file <- factor(loan$previous_loan_defaults_on_file)
loan$person_gender <- factor(loan$person_gender)
loan$person_education <- factor(loan$person_education)
loan$person_home_ownership <- factor(loan$person_home_ownership)
loan$loan_intent <- factor(loan$loan_intent)

ggplot(loan, aes(x = person_income)) +
  geom_histogram(bins = 30) +
  labs(title = "Histrogram of Income")
```



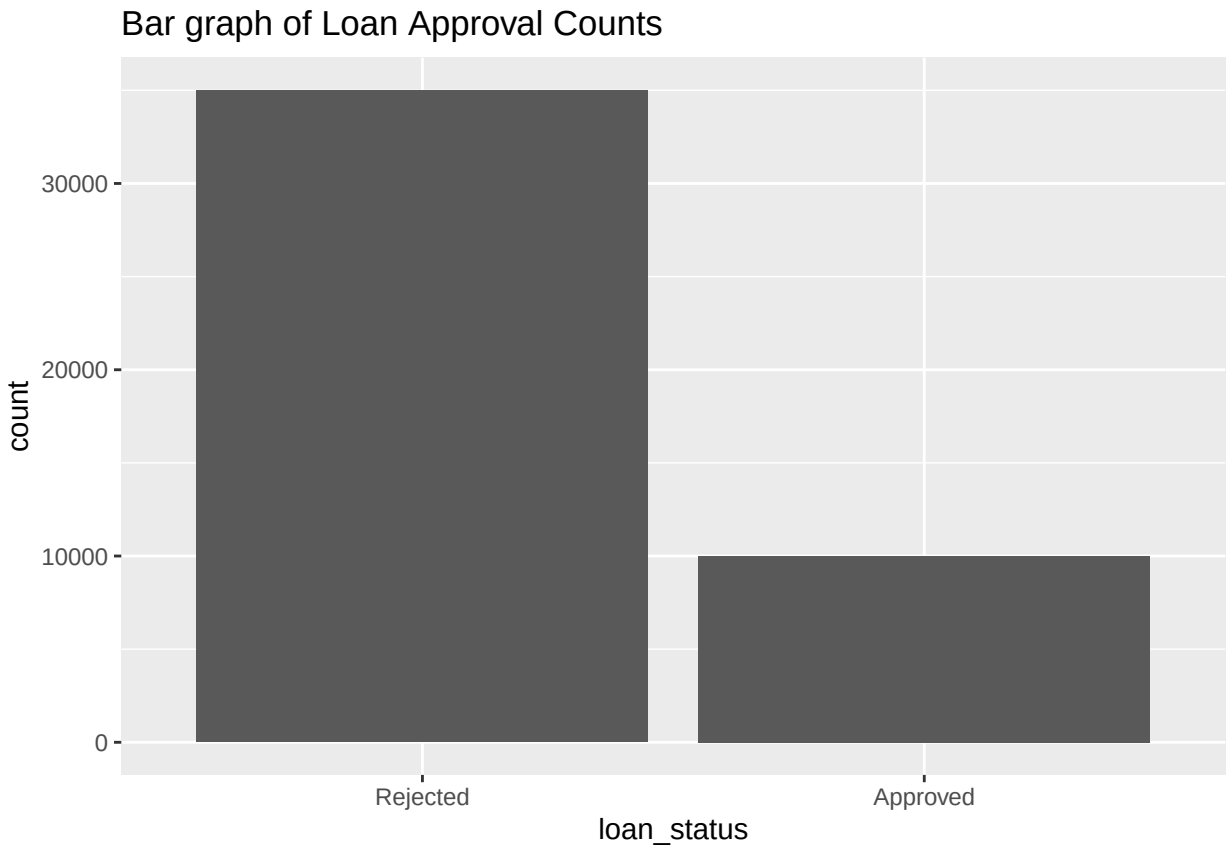

```r
ggplot(loan, aes(x = credit_score)) +
  geom_histogram(bins = 30) +
  labs(title = "Histrogram Credit Score")
```



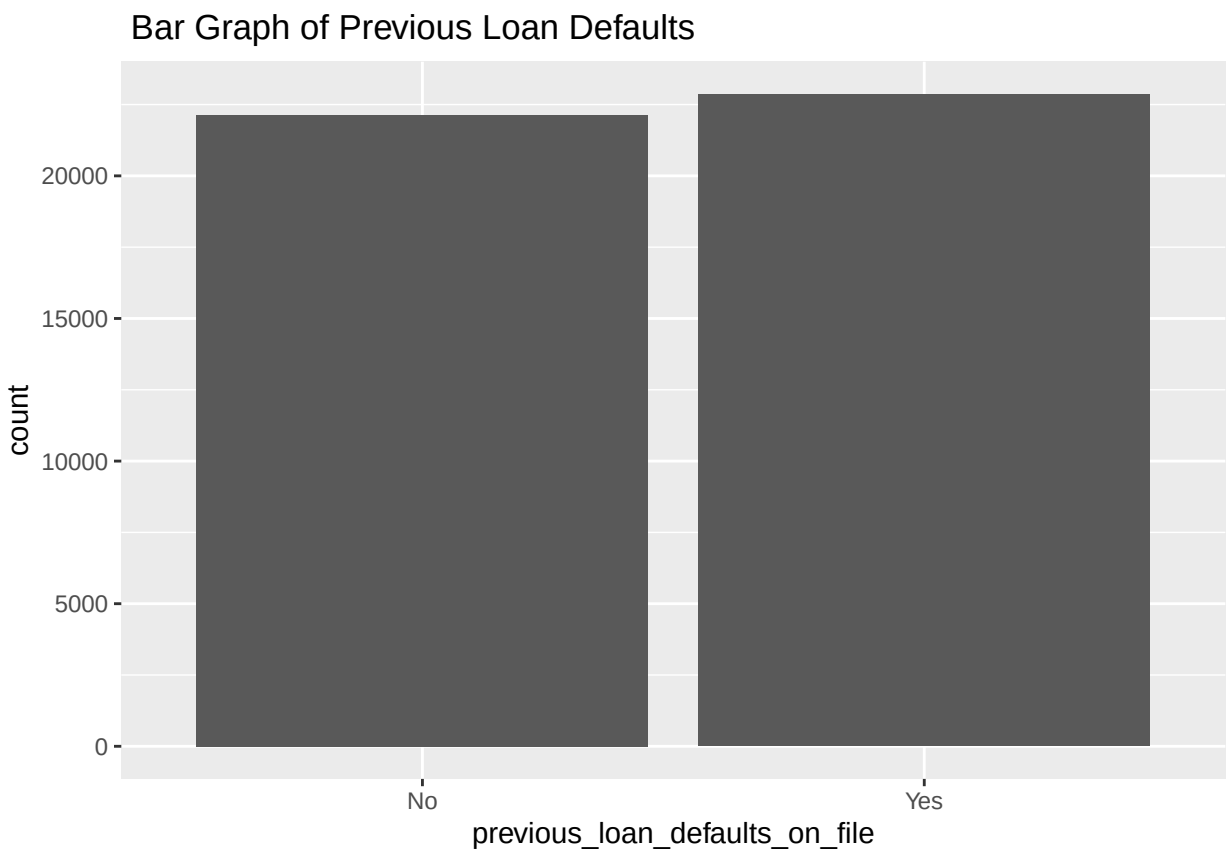

```r
ggplot(loan, aes(x = loan_percent_income)) +
  geom_histogram(bins = 30) +
  labs(title = "Histrogram of Loan Percent Income")
```



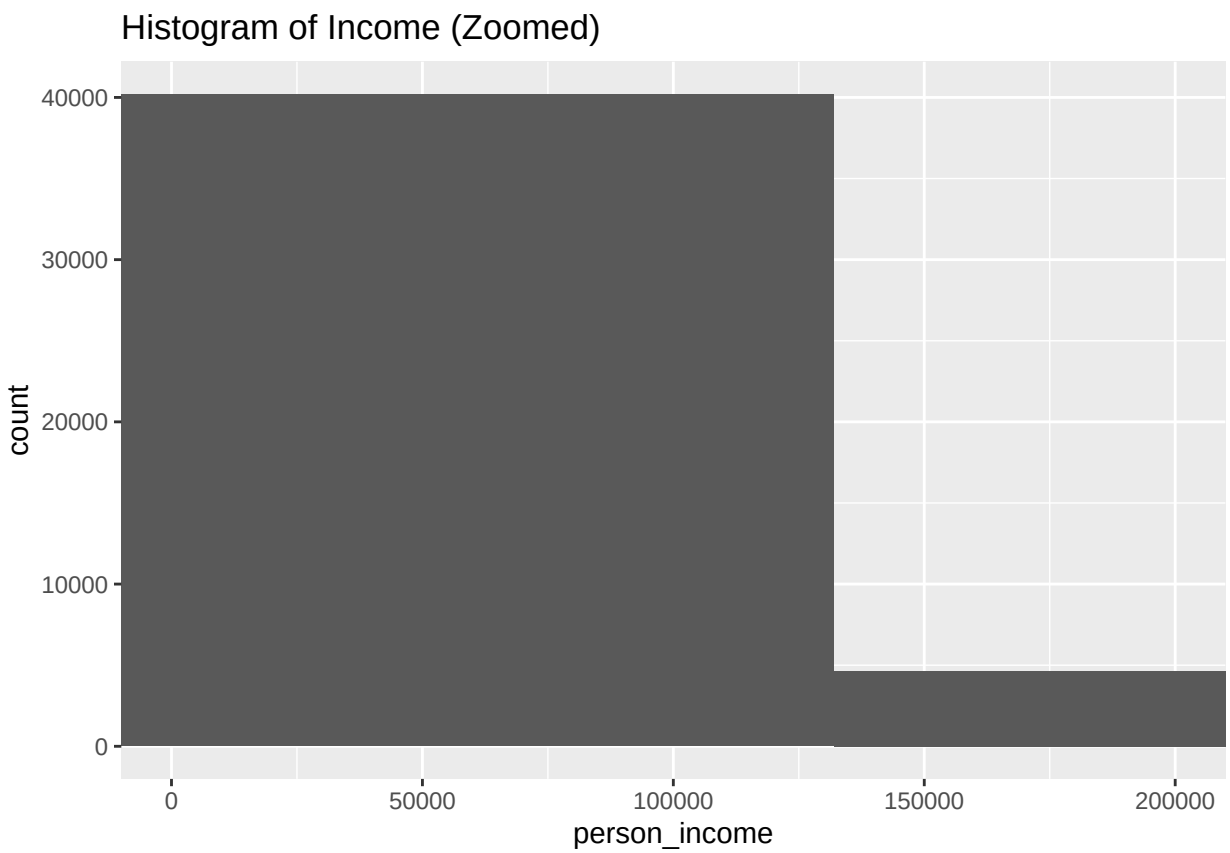

```
ggplot(loan, aes(x = loan_status)) +
  geom_bar() +
  labs(title = "Bar graph of Loan Approval Counts")
```




```
ggplot(loan, aes(x = previous_loan_defaults_on_file)) +
  geom_bar() +
  labs(title = " Bar Graph of Previous Loan Defaults")
```

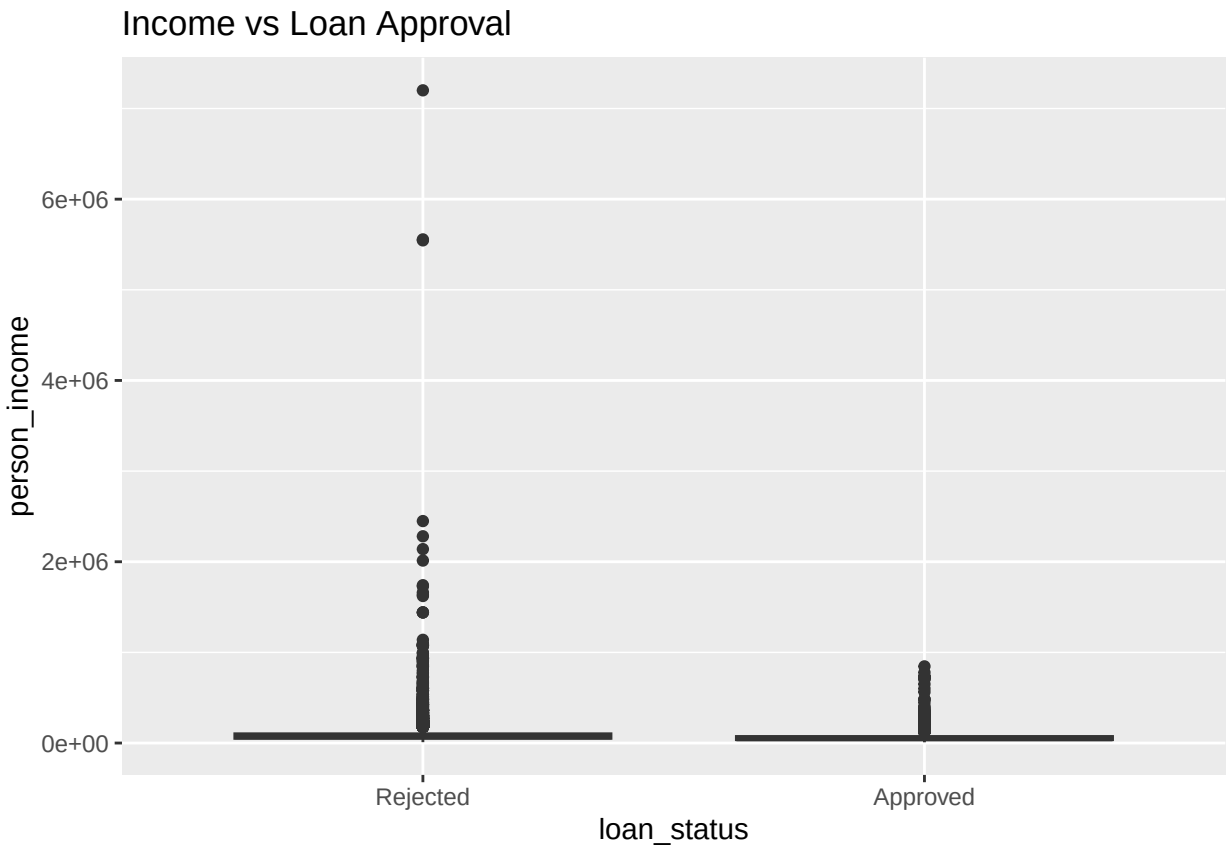



```r
ggplot(loan, aes(x = person_income)) +
  geom_histogram(bins = 30) +
  coord_cartesian(xlim = c(0, 200000)) +
  labs(title = "Histogram of Income (Zoomed)")
```

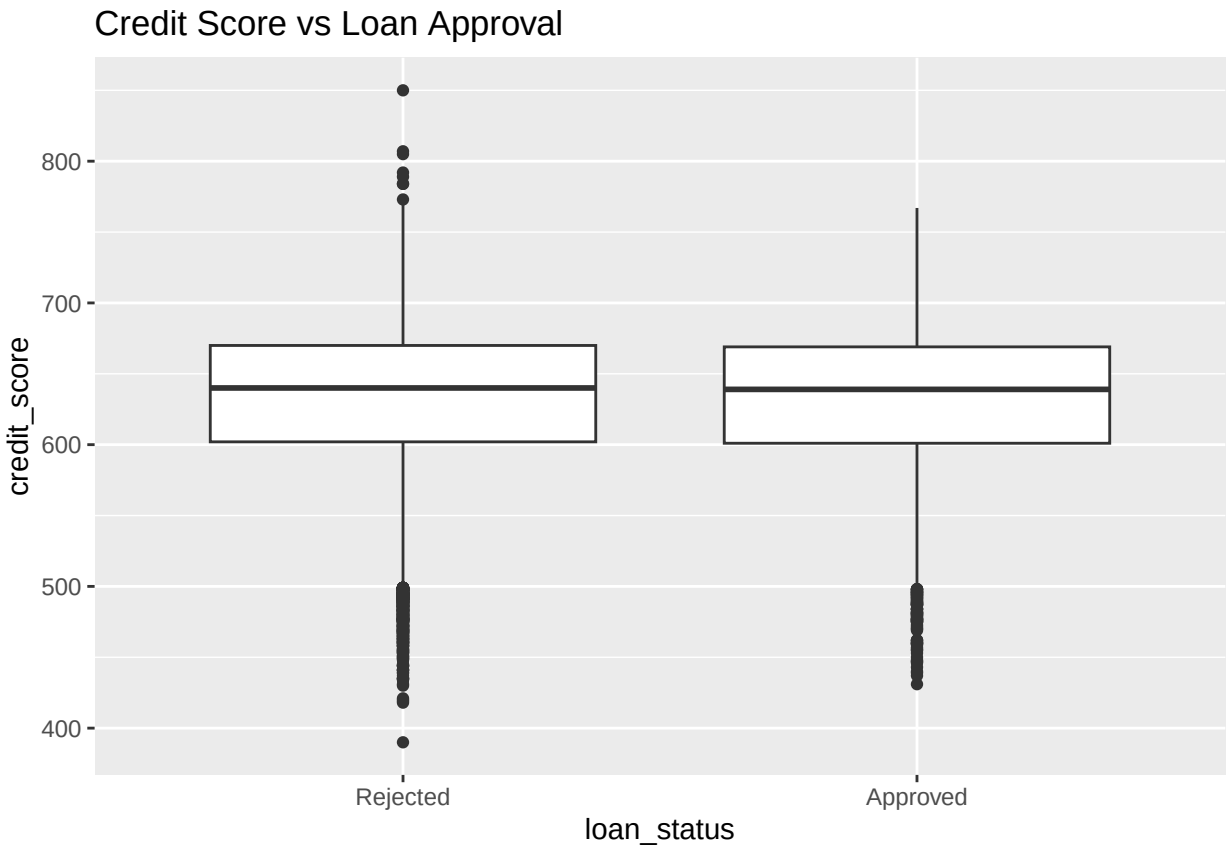


Multivariate EDA

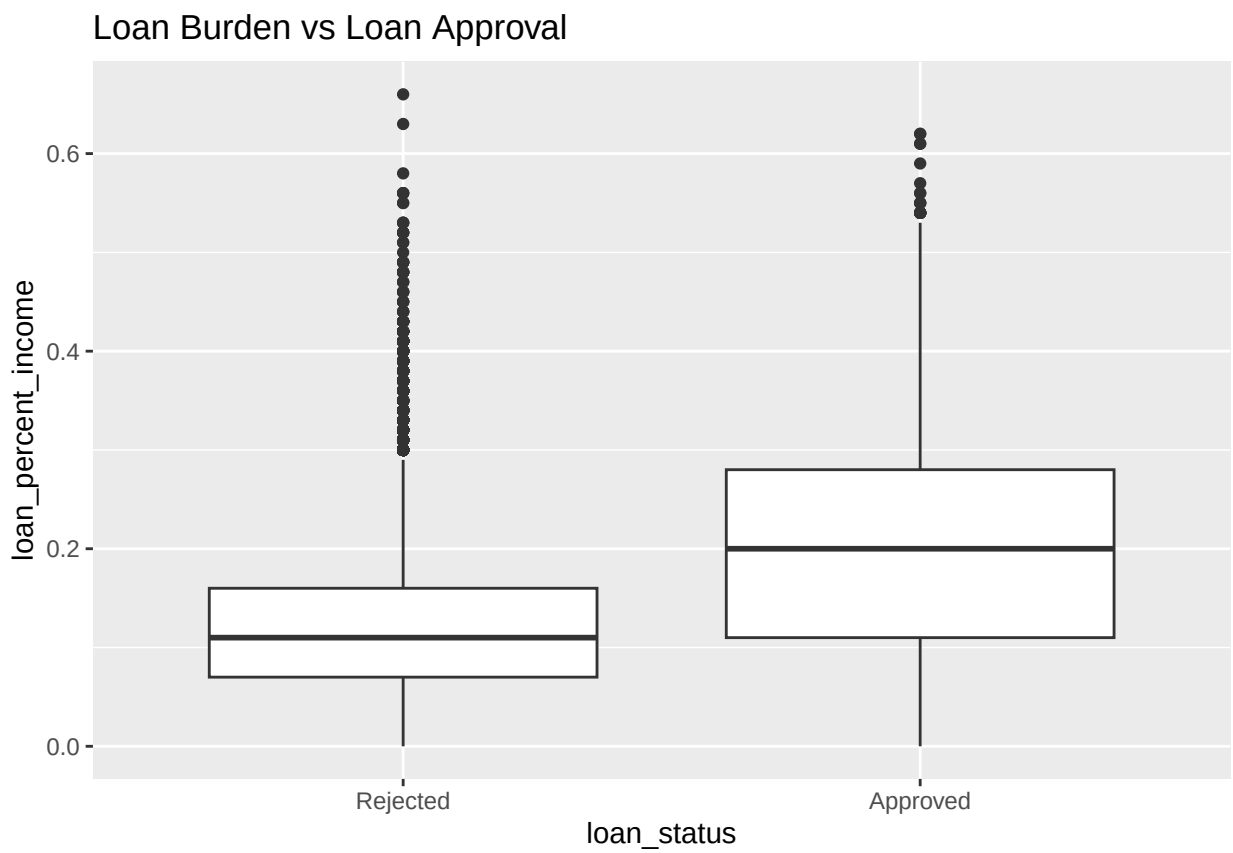
```
ggplot(loan, aes(x = loan_status, y = person_income)) +
  geom_boxplot() +
  labs(title = "Income vs Loan Approval")
```



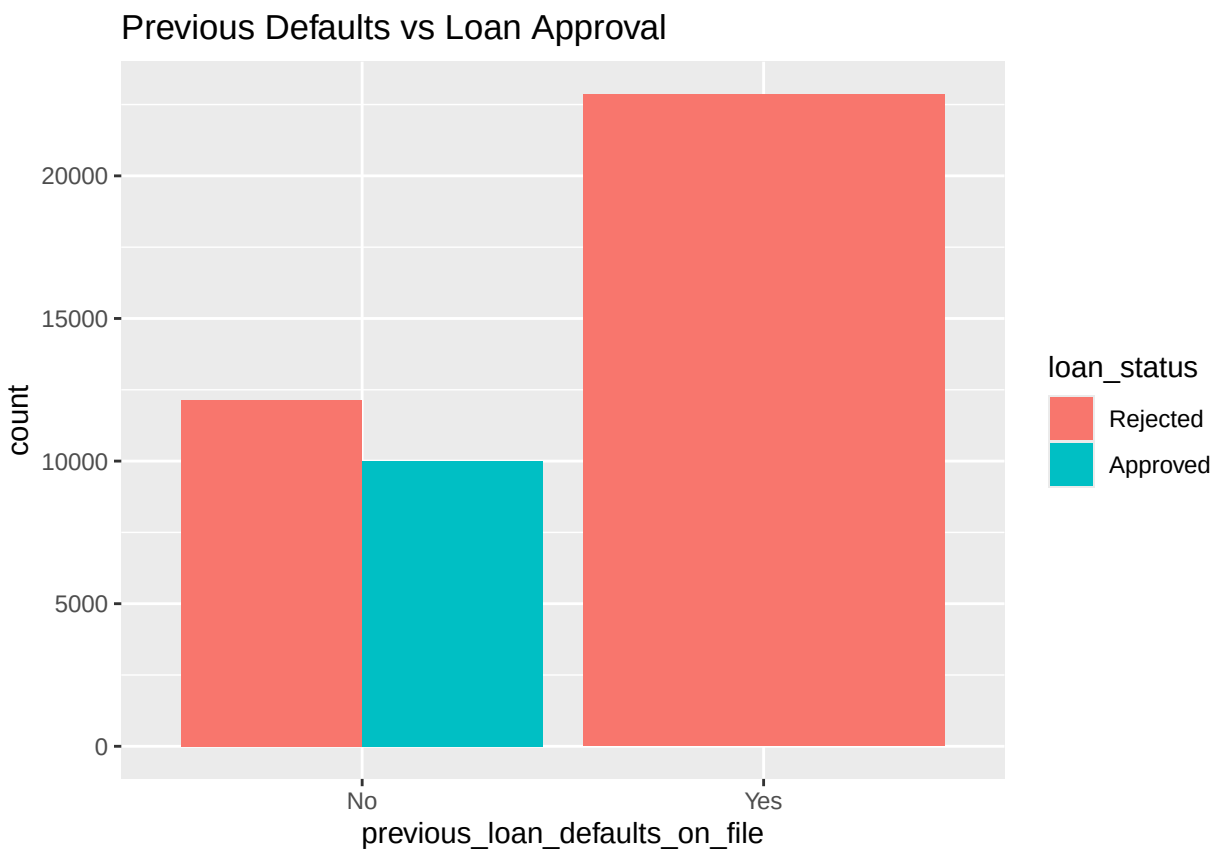

```
ggplot(loan, aes(x = loan_status, y = credit_score)) +
  geom_boxplot() +
  labs(title = "Credit Score vs Loan Approval")
```




```
ggplot(loan, aes(x = loan_status, y = loan_percent_income)) +
  geom_boxplot() +
  labs(title = "Loan Burden vs Loan Approval")
```

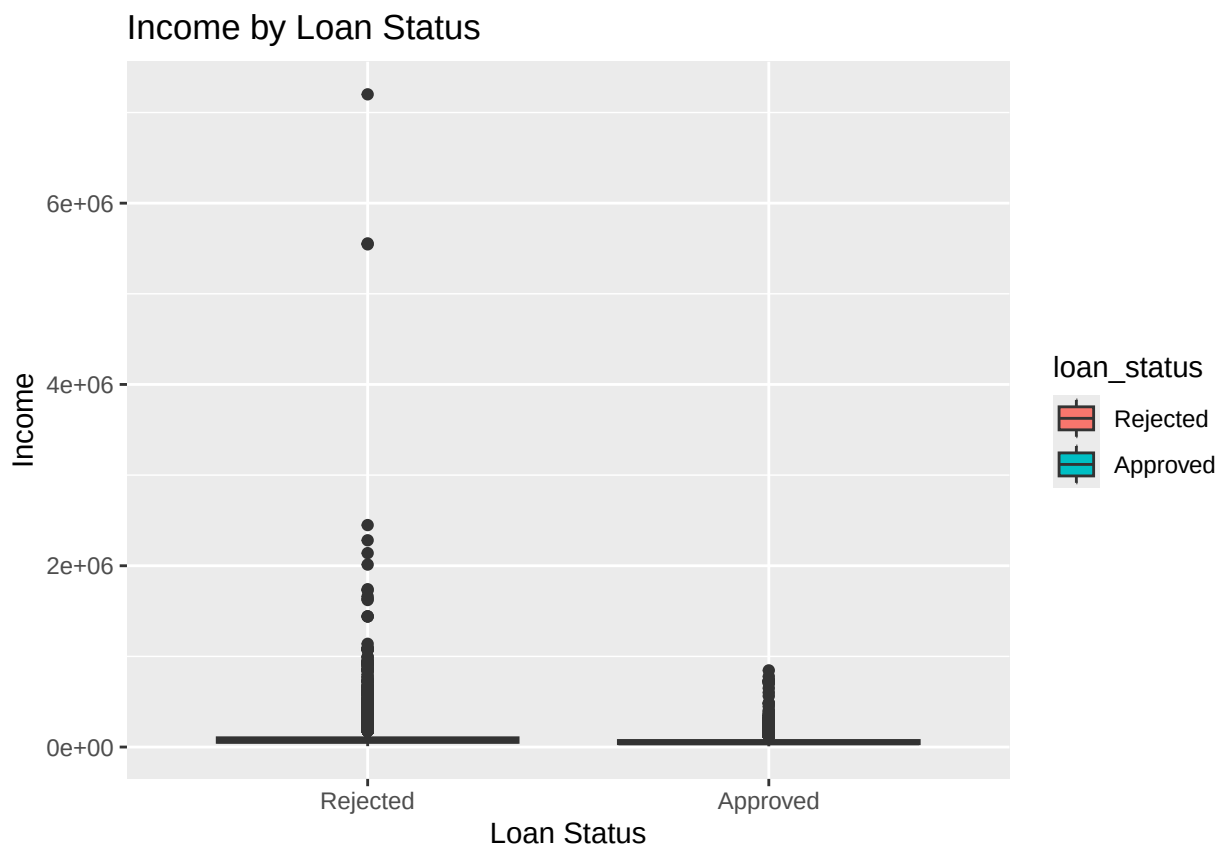



```
ggplot(loan, aes(x = previous_loan_defaults_on_file, fill = loan_status)) +
  geom_bar(position = "dodge") +
  labs(title = "Previous Defaults vs Loan Approval")
```

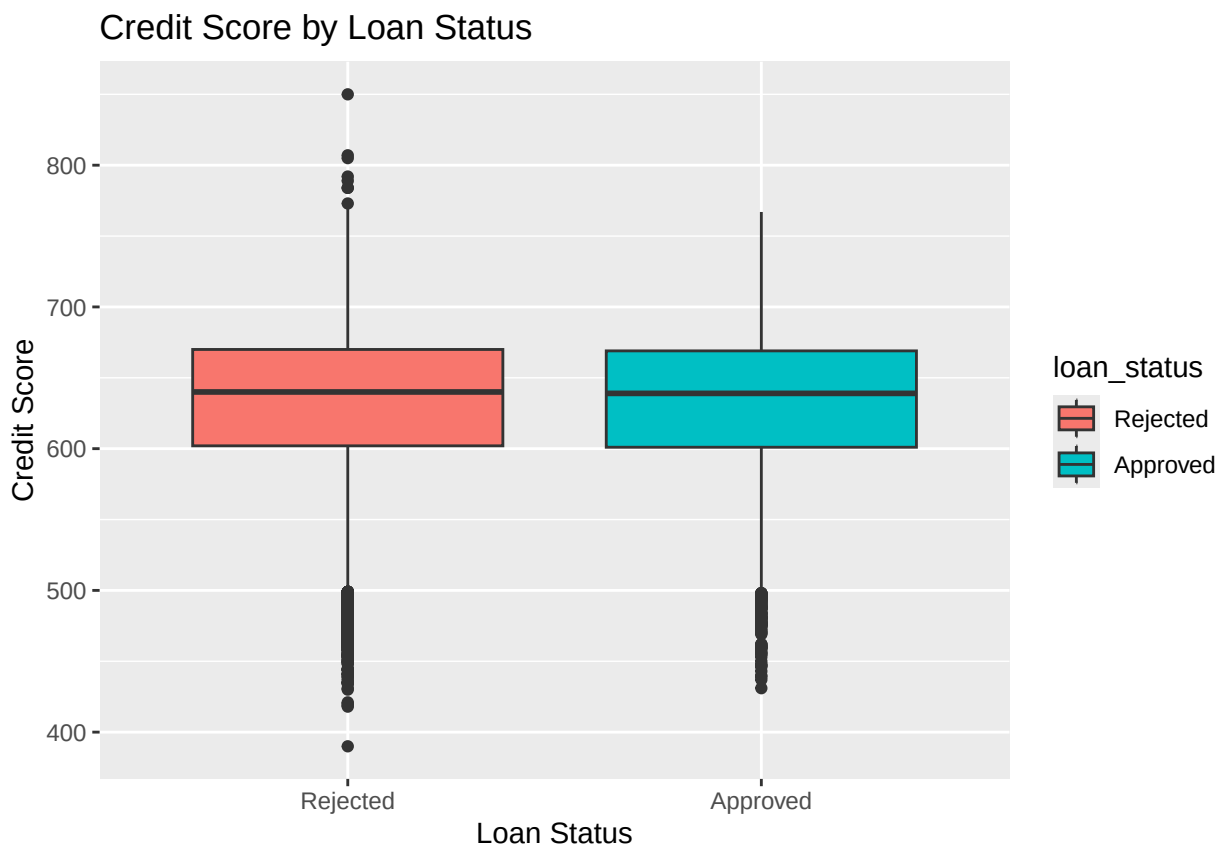


Numeric vs loan status predictors

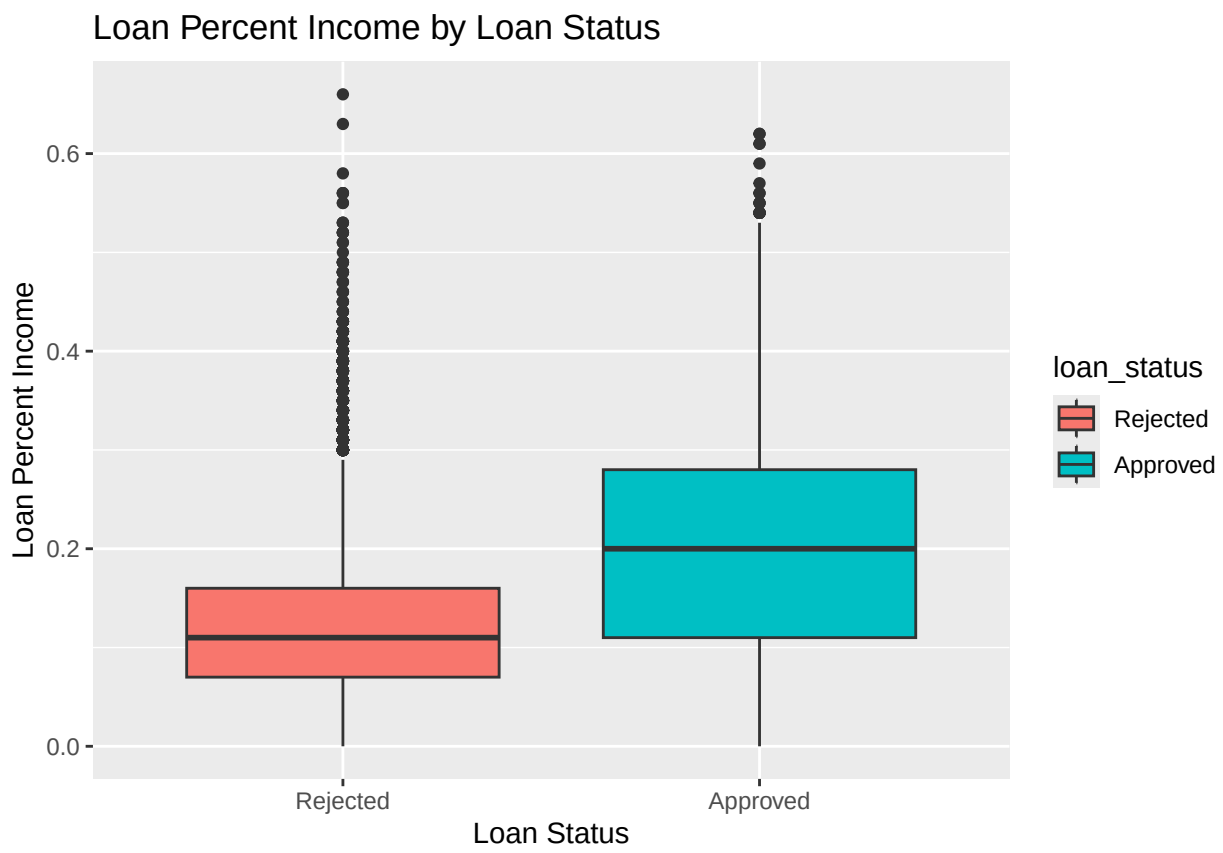
```
library(ggplot2)

ggplot(loan, aes(x = loan_status, y = person_income, fill = loan_status)) +
  geom_boxplot() +
  labs(title = "Income by Loan Status",
       x = "Loan Status", y = "Income")
```




```r
ggplot(loan, aes(x = loan_status, y = credit_score, fill = loan_status)) +
  geom_boxplot() +
  labs(title = "Credit Score by Loan Status",
       x = "Loan Status", y = "Credit Score")
```

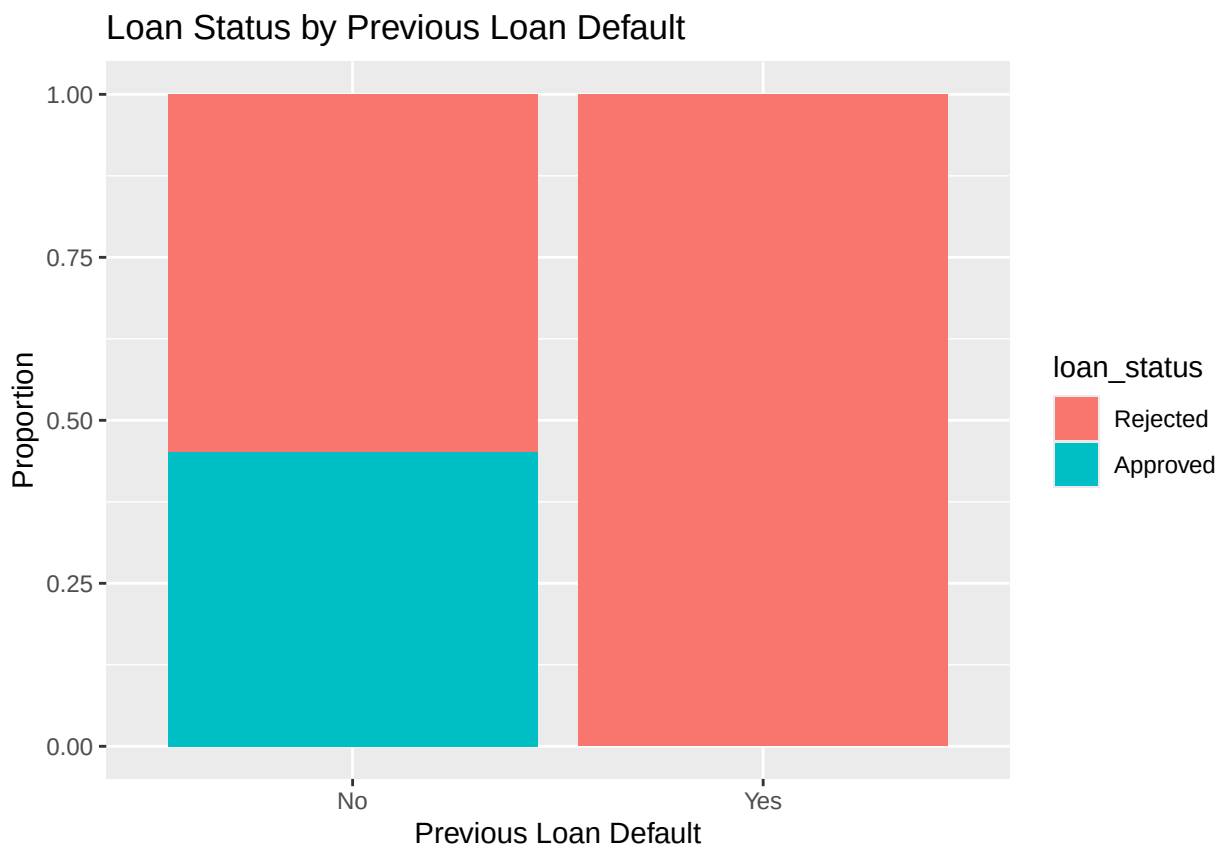



```
ggplot(loan, aes(x = loan_status, y = loan_percent_income, fill = loan_status)) +
  geom_boxplot() +
  labs(title = "Loan Percent Income by Loan Status",
       x = "Loan Status", y = "Loan Percent Income")
```

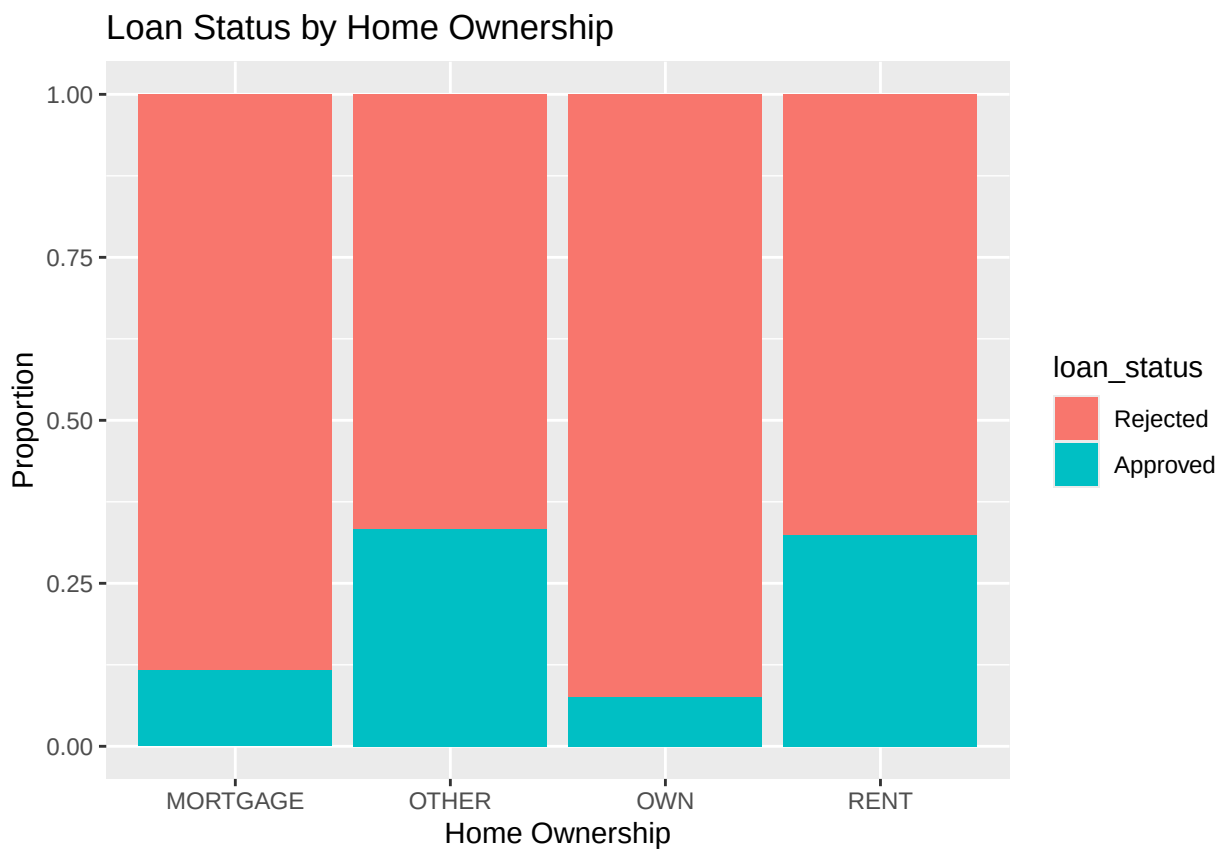


caterogical vs loan status

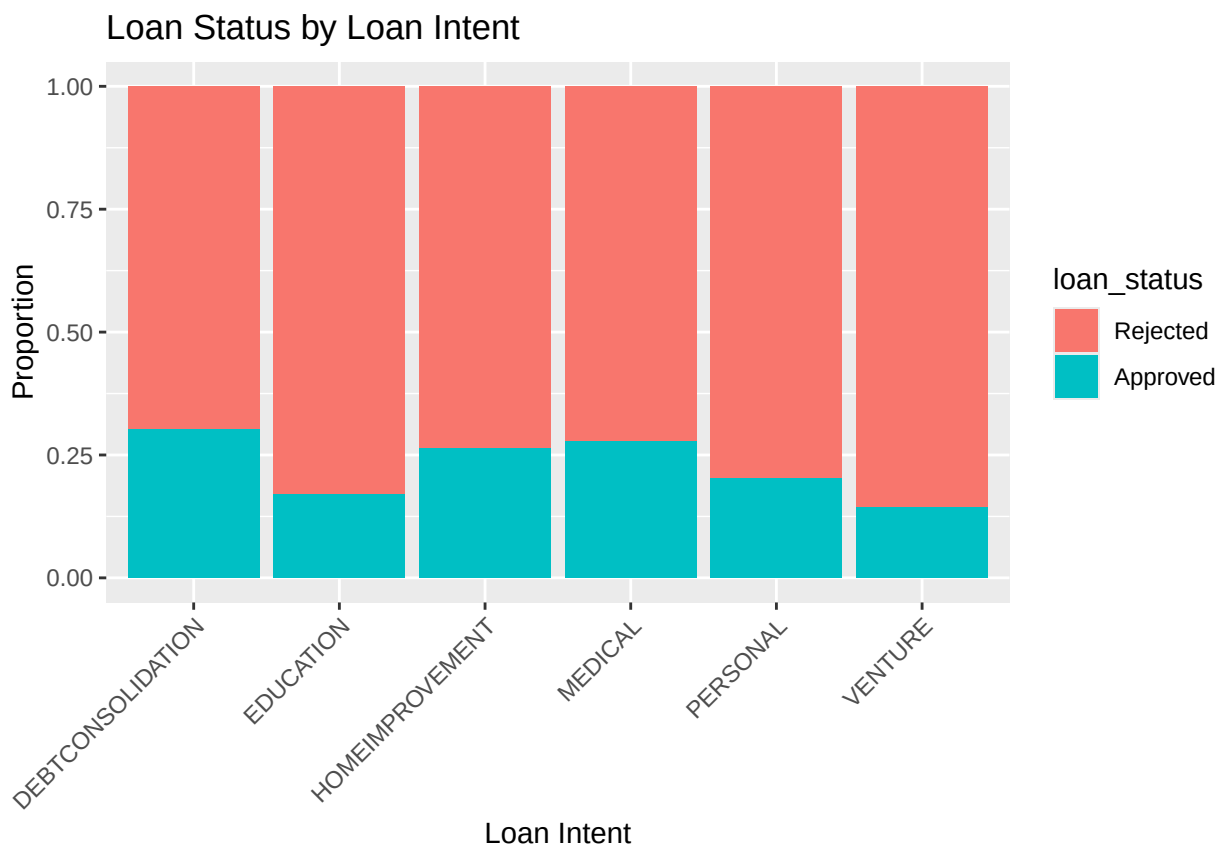
```
ggplot(loan, aes(x = previous_loan_defaults_on_file, fill = loan_status)) +
  geom_bar(position = "fill") +
  labs(title = "Loan Status by Previous Loan Default",
       x = "Previous Loan Default", y = "Proportion")
```




```r
ggplot(loan, aes(x = person_home_ownership, fill = loan_status)) +
  geom_bar(position = "fill") +
  labs(title = "Loan Status by Home Ownership",
       x = "Home Ownership", y = "Proportion")
```




```r
ggplot(loan, aes(x = loan_intent, fill = loan_status)) +
  geom_bar(position = "fill") +
  labs(title = "Loan Status by Loan Intent",
       x = "Loan Intent", y = "Proportion") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Multicollinearity

```
numeric_vars <- loan[, c("person_age", "person_income", "loan_amnt",
                         "loan_int_rate", "credit_score", "loan_percent_income")]

cor(numeric_vars, use = "complete.obs")
```

```
##                      person_age person_income   loan_amnt loan_int_rate
## person_age           1.00000000   0.193697781 0.050749541   0.013401640
## person_income        0.19369778   1.000000000 0.242290131   0.001509828
## loan_amnt            0.05074954   0.242290131 1.000000000   0.146093082
## loan_int_rate        0.01340164   0.001509828 0.146093082   1.000000000
## credit_score         0.17843247   0.035919225 0.009074282   0.011497752
## loan_percent_income -0.04329864  -0.234176548 0.593011449   0.125209488
##                     credit_score loan_percent_income
## person_age           0.178432470         -0.04329864
## person_income        0.035919225         -0.23417655
## loan_amnt            0.009074282          0.59301145
## loan_int_rate        0.011497752          0.12520949
## credit_score         1.000000000         -0.01148310
## loan_percent_income -0.011483096          1.00000000
```

Logistic Regression

```r
model1 <- glm(loan_status ~ credit_score + person_income +
              loan_percent_income + loan_int_rate +
              loan_amnt,
              data = loan, family = binomial)

summary(model1)
```

```
##
## Call:
## glm(formula = loan_status ~ credit_score + person_income + loan_percent_income +
##     loan_int_rate + loan_amnt, family = binomial, data = loan)
##
## Coefficients:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)         -6.361e+00  1.820e-01 -34.957   <2e-16 ***
## credit_score        -3.167e-04  2.655e-04  -1.193    0.233
## person_income        3.020e-07  1.969e-07   1.534    0.125
## loan_percent_income  1.543e+01  2.222e-01  69.436   <2e-16 ***
## loan_int_rate        3.351e-01  5.121e-03  65.427   <2e-16 ***
## loan_amnt           -1.078e-04  3.079e-06 -35.000   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 47674  on 44999  degrees of freedom
## Residual deviance: 35175  on 44994  degrees of freedom
## AIC: 35187
##
## Number of Fisher Scoring iterations: 6
```

```r
model2 <- glm(loan_status ~ credit_score + person_income +
              loan_percent_income + loan_int_rate +
              loan_amnt + previous_loan_defaults_on_file,
              data = loan, family = binomial)

summary(model2)
```

```
##
## Call:
## glm(formula = loan_status ~ credit_score + person_income + loan_percent_income +
##     loan_int_rate + loan_amnt + previous_loan_defaults_on_file,
##     family = binomial, data = loan)
##
## Coefficients:
##                                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)                        1.719e-01  2.477e-01   0.694    0.488
## credit_score                      -8.927e-03  3.746e-04 -23.828   <2e-16 ***
## person_income                      3.197e-07  2.112e-07   1.513    0.130
## loan_percent_income                1.557e+01  2.842e-01  54.786   <2e-16 ***
```

```
## loan_int_rate                      3.353e-01  6.283e-03  53.366   <2e-16 ***
## loan_amnt                         -1.049e-04  3.666e-06 -28.616   <2e-16 ***
## previous_loan_defaults_on_fileYes -2.058e+01  1.041e+02  -0.198    0.843
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 47674  on 44999  degrees of freedom
## Residual deviance: 21417  on 44993  degrees of freedom
## AIC: 21431
##
## Number of Fisher Scoring iterations: 19
```

```r
AIC(model1, model2)
```

```
##        df      AIC
## model1  6 35186.87
## model2  7 21430.89
```

final model

```r
model_final <- glm(loan_status ~ credit_score + person_income +
                   loan_percent_income + loan_int_rate +
                   loan_amnt,
                   data = loan, family = binomial)

summary(model_final)
```

```
##
## Call:
## glm(formula = loan_status ~ credit_score + person_income + loan_percent_income +
##     loan_int_rate + loan_amnt, family = binomial, data = loan)
##
## Coefficients:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)         -6.361e+00  1.820e-01 -34.957   <2e-16 ***
## credit_score        -3.167e-04  2.655e-04  -1.193    0.233
## person_income        3.020e-07  1.969e-07   1.534    0.125
## loan_percent_income  1.543e+01  2.222e-01  69.436   <2e-16 ***
## loan_int_rate        3.351e-01  5.121e-03  65.427   <2e-16 ***
## loan_amnt           -1.078e-04  3.079e-06 -35.000   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 47674  on 44999  degrees of freedom
## Residual deviance: 35175  on 44994  degrees of freedom
## AIC: 35187
##
## Number of Fisher Scoring iterations: 6
```